GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

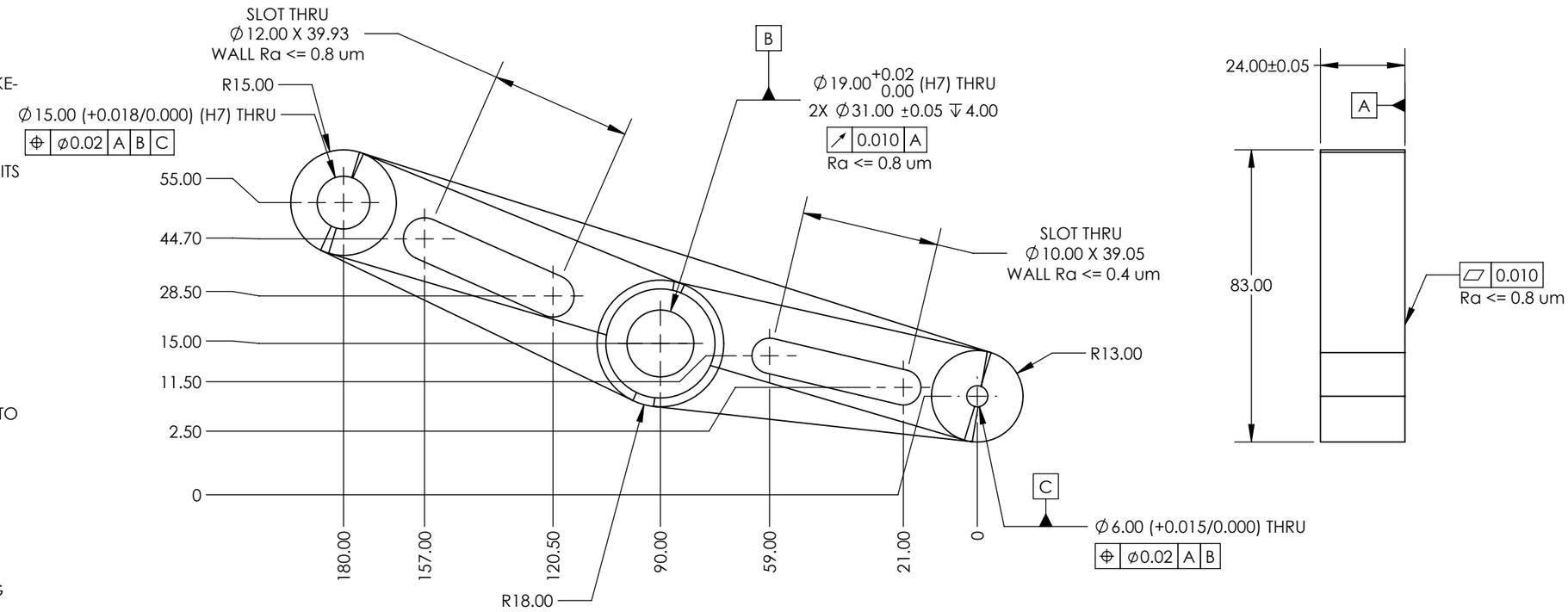
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET GOVERNS NOMINAL GEOMETRY: W3-BIKE-ROCKER-8.3.STEP.
1.3 GENERAL TOLERANCES: LINEAR ISO 2768-m (±0.20 mm), ANGULAR ±0.1 DEG.
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM A).

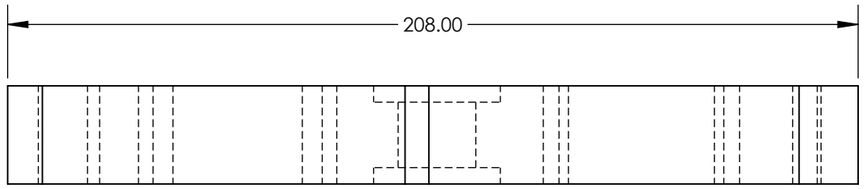
2. RAW MATERIAL & METALLURGY SPECIFICATION:
2.1 MATERIAL: PRE-STRETCHED FORGED AIRCRAFT ALUMINUM ALLOY 7075-T651 PER AMS 4045 / ASTM B209.
2.2 MECHANICAL PROPERTIES: MINIMUM TENSILE YIELD STRENGTH Rp0.2 >= 505 MPa, ULTIMATE Rm >= 570 MPa, ELONGATION >= 11%.
2.3 100% ULTRASONIC TESTED TO CLASS A PER AMS-STD-2154 TO GUARANTEE VOID-FREE INTERNAL STRUCTURE.

3. 5-AXIS CNC MACHINING & GD&T CONTROLS:
3.1 PROCESS ROUTE: 2-SETUP FIXTURING ON 5-AXIS CNC MACHINING CENTER:
- OP10: TOP/LATERAL ROUGHING, POCKET MILLING, AND HOLE PRE-MACHINING.
- OP20: FLIP ON CUSTOM CONTOURED SOFT JAWS, FACING AND FINISH BORING OF BEARING SEATS.
3.2 PRIMARY DATUM [A] (BOTTOM BASE): FLATNESS <= 0.010 mm ACROSS CONTACT ENVELOPE. SURFACE FINISH Ra <= 0.8 µm.
3.3 BEARING BORES Ø19.000 mm (DATUM B) AND Ø15.000 mm: TOLERANCE CLASS H7 (+0.018/-0.000 mm). CYLINDRICITY <= 0.006 mm. SURFACE FINISH Ra <= 0.4 µm.
3.4 BEARING SEAT RECESSES: DEPTH 4.00 ±0.03 mm, AXIAL RUNOUT <= 0.010 mm RELATIVE TO DATUM [B].
3.5 2X SLOT THRU POCKETS: WALL THICKNESS RELATIVE TO BEARING BOSS [B] MUST MAINTAIN MINIMUM >= 8.00 mm SOLID LIGAMENT TO PREVENT BORE OVALIZATION UNDER 5.4 kN DYNAMIC LOAD.

4. SURFACE TREATMENT & DEBURRING:
4.1 100% DEBURR ALL EDGES: CORNER RADII/CHAMFERS R0.20-R0.40 mm UNDER MICROSCOPE INSPECTION. ZERO SHARP EDGES OR TOOL BURRS PERMITTED.
4.2 SURFACE FINISHING: FINE GLASS BEAD BLAST (120 MESH) FOLLOWED BY MIL-A-8625F TYPE II CLASS 2 ANODIZATION (MATTE BLACK, MINIMUM COATING THICKNESS 25 µm).

5. QUALITY VERIFICATION & CMM AUDIT:
5.1 100% CMM INSPECTION FOR PIVOT CENTER DISTANCES:
- P3 TO P1: 98.49 ±0.02 mm.
- P1 TO P2: 91.24 ±0.02 mm.
- P3 TO P2: 188.22 ±0.03 mm.
5.2 CLEANLINESS: ULTRASONIC CLEAN IN ISOPROPYL ALCOHOL (IPA), INDIVIDUALLY PACKED IN PROTECTIVE CORROSION-INHIBITING VCI POUCHES.







For reference

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ISO 2768-m (±0.20 mm), Angular ±0.1°	FINISH: Bead Blast (120 Mesh) + MIL-A-8625F Type II Class 2 Hard Anodize (Matte Black, 25 µm), Ra <= 0.8 µm on Pockets				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION: A
						Mode 3: Simultaneous 5-Axis Subtractive CNC Milling (Single-Billet Machined)
	NAME	SIGNATURE	DATE			TITLE: 5-AXIS CNC MTB SUSPENSION UPPER ROCKER LINKAGE
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Forged Pre-Stretched Aircraft Aluminum Alloy 7075-T651 (AMS 4045)		DWG NO. W3-BIKE-ROCKER-8.3 / A3
				WEIGHT: ~182 g		SCALE NTS / SHEET 1 OF 1